

HOSPITAL QUALITY & READMISSION ANALYSIS

Exploring Organizational and Patient Experience Factors Associated with Hospital Performance

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What This Deck Covers

01

The Story

Why hospital readmissions matter and what three CMS datasets can tell us

02

The Data

Merging Hospital General Information, HCAHPS, and the Readmissions Reduction Program

03

10 Research Questions

Each with a publication-ready Plotly visual, statistical test, and takeaway

04

Synthesis

What actually predicts excess readmissions, and which hospitals get it right

05

Live Dashboard

An interactive Streamlit companion for exploring the findings yourself

Why Hospital Readmissions?

Hospital readmission rates are one of healthcare's clearest quality signals they reflect clinical care, discharge planning, and care coordination all at once. Excessive readmissions raise costs and often point to gaps in patient care.

This project integrates three CMS Care Compare datasets to ask: which organizational and patient-experience factors are actually associated with better readmission performance?

3,055

hospitals analyzed

51

U.S. states

6

clinical conditions

3

CMS datasets merged

Data & Methodology

Three CMS Care Compare datasets, merged on Facility ID



Hospital General Information

Ownership, hospital type, emergency services, CMS overall rating, and counts of mortality/safety/readmission measures beating the national average.



HCAHPS Patient Survey

8 patient-experience measures (communication, cleanliness, quietness, recommend, overall rating) combined into a Patient Experience Index.



Readmissions Reduction Program

Excess Readmission Ratio, predicted vs. expected rates, and discharge/readmission counts across 6 major conditions.

Analysis-ready integrated dataset

18,330

rows (hospital × condition)

26

merged columns

3,055

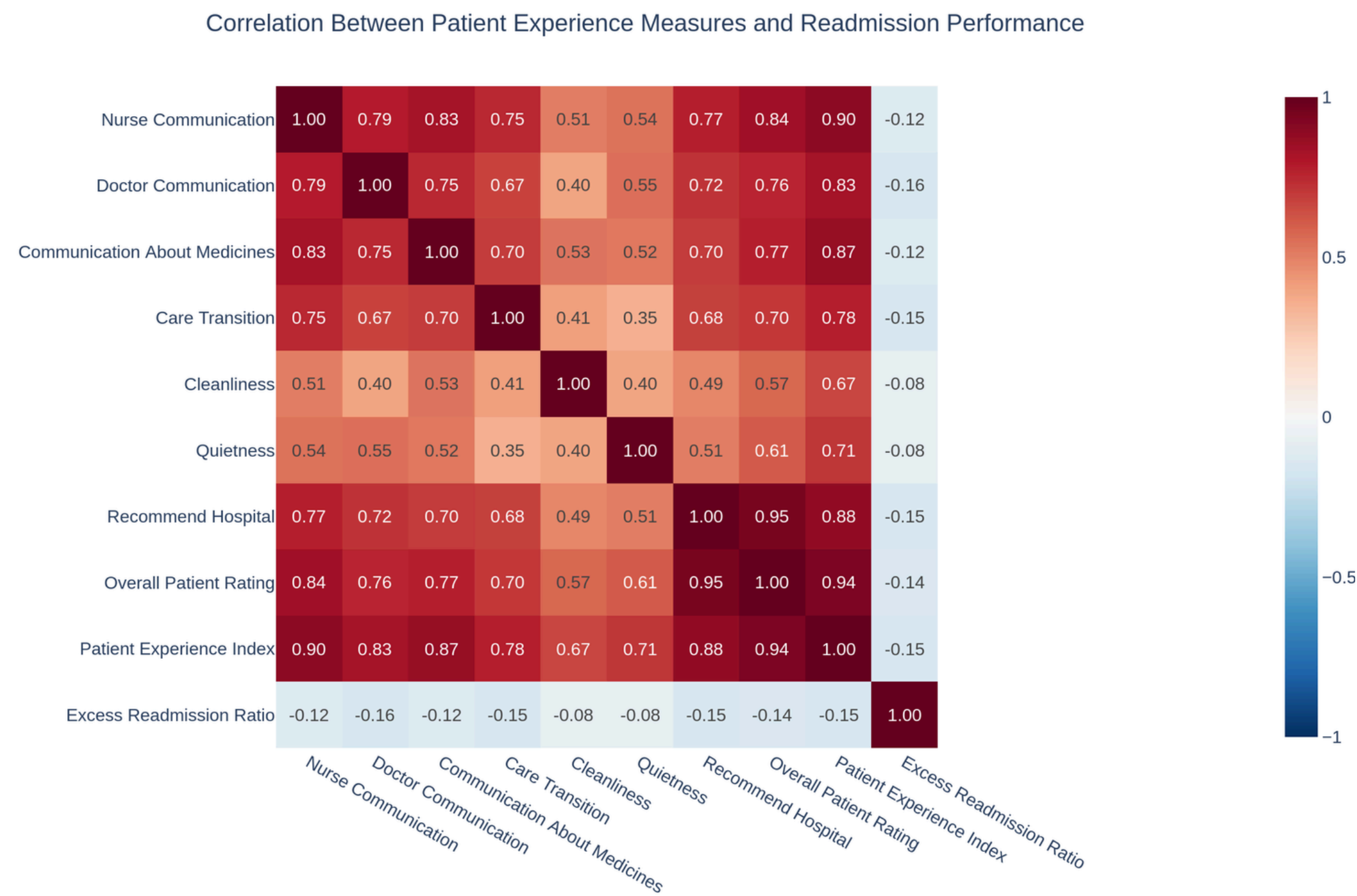
unique hospitals

51

states

Which patient-experience dimensions relate most to readmissions?

Correlation matrix — 9 patient-experience measures vs. Excess Readmission Ratio



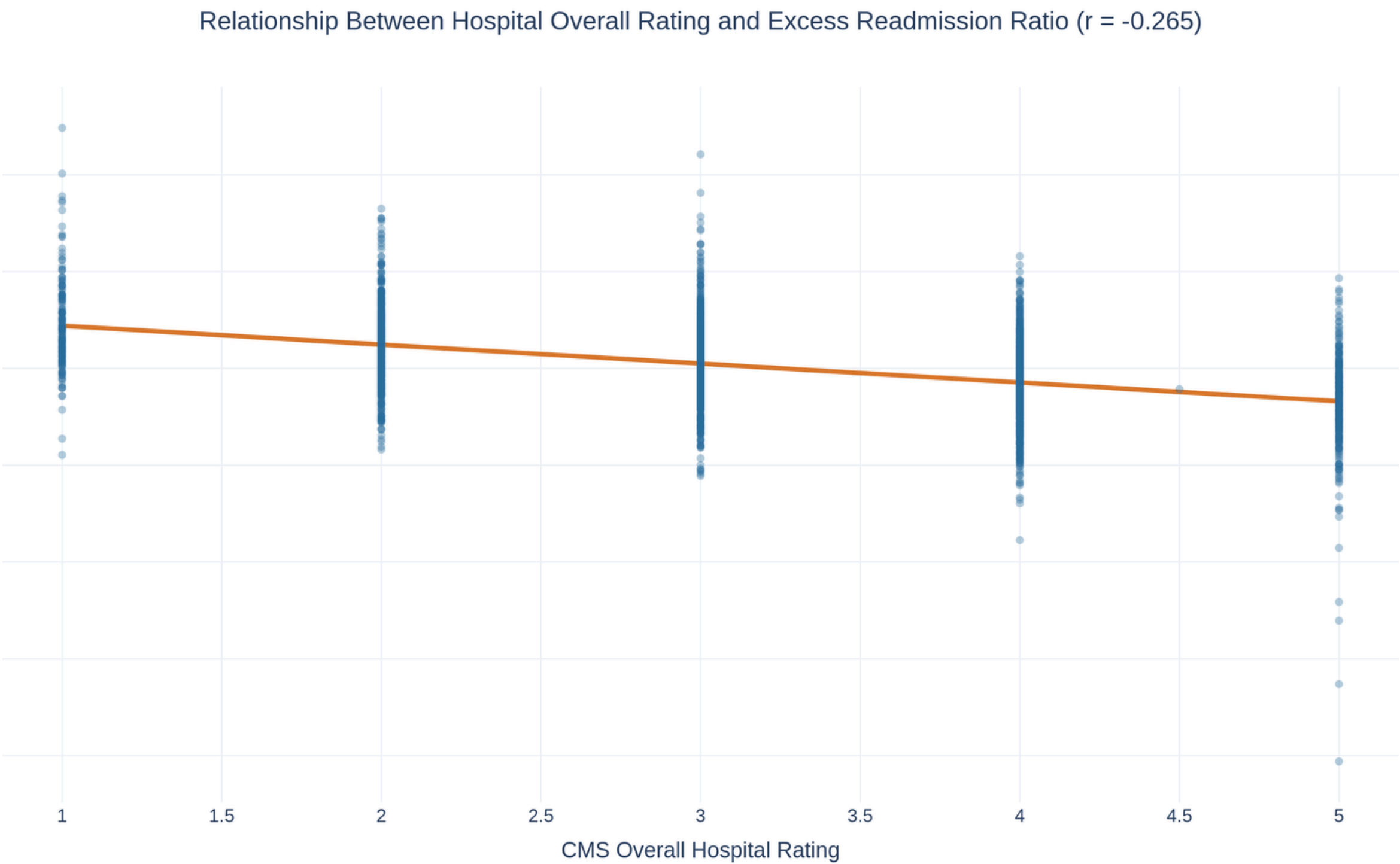
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Key Takeaway

All experience measures correlate negatively with excess readmissions, but weakly ($r = -0.08$ to -0.16). Doctor Communication ($r = -0.16$) and Care Transition ($r = -0.15$) show the strongest links clear communication and discharge planning matter more than environmental factors like cleanliness or quietness (both $r = -0.08$).

Do higher CMS ratings mean fewer excess readmissions?

3,055 hospitals · Pearson $r = -0.265$



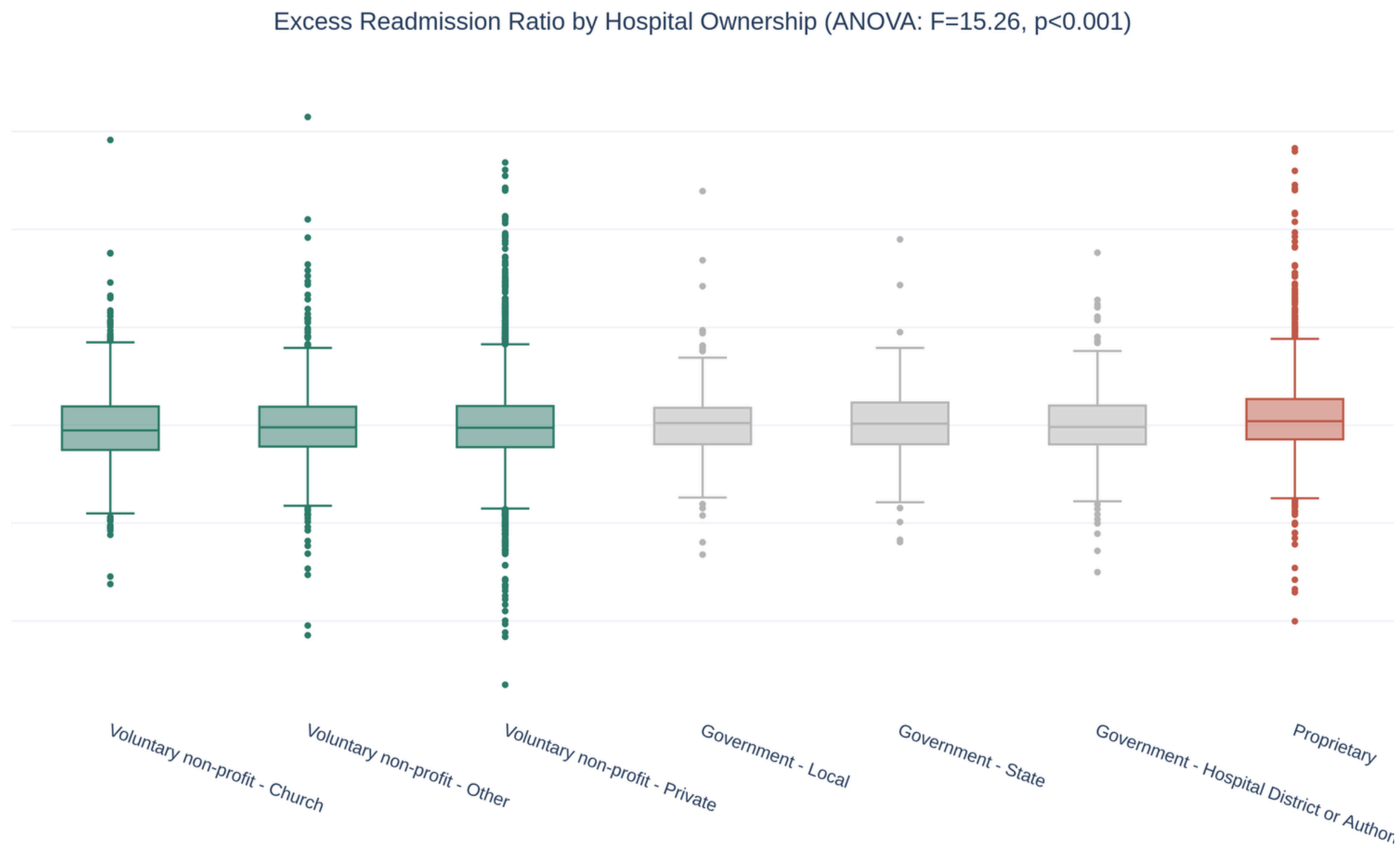
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Key Takeaway

A weak negative relationship: better-rated hospitals tend toward lower excess readmission ratios, and the OLS trend line confirms the direction. But the wide scatter around the line shows rating alone explains only part of the story other organizational and clinical factors are clearly at play.

Does ownership structure affect readmission performance?

One-way ANOVA: $F = 15.26, p < 0.001$



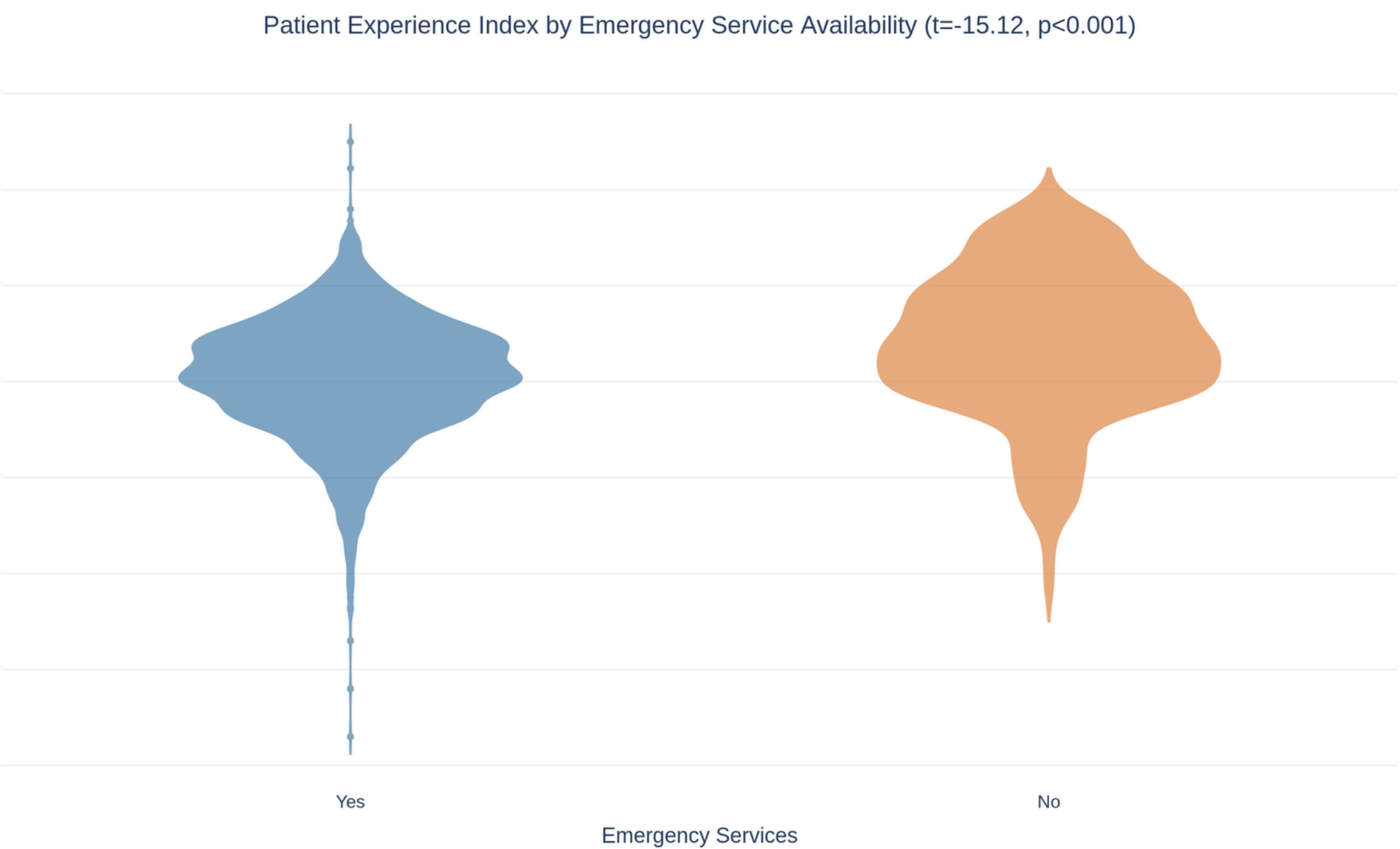
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Key Takeaway

Voluntary Non-profit (Church) hospitals post the lowest average Excess Readmission Ratio (0.99); Proprietary hospitals the highest (1.02). Differences are statistically significant but modest in size ownership matters, but isn't the dominant driver of performance.

Do ER hospitals deliver a different patient experience?

Welch's t-test: $t = -15.12, p < 0.001$



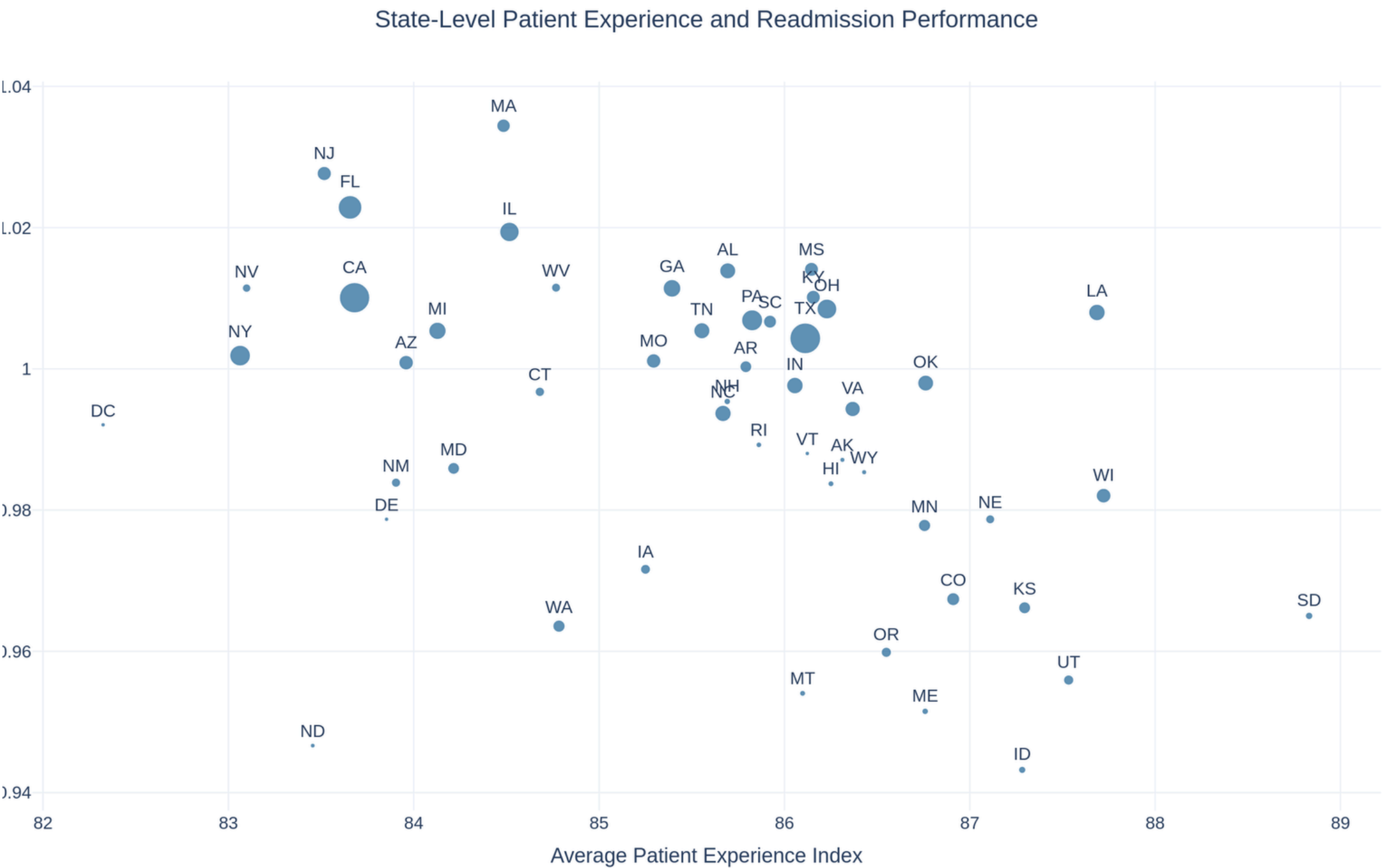
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Key Takeaway

Hospitals without emergency services score higher on Patient Experience (86.97 vs. 85.18) a small but significant gap. Higher patient volume and case complexity in ER hospitals likely affects perceived communication and wait times, not necessarily care quality itself.

Which states balance great experience with low readmissions?

State-level averages · bubble size = number of hospitals



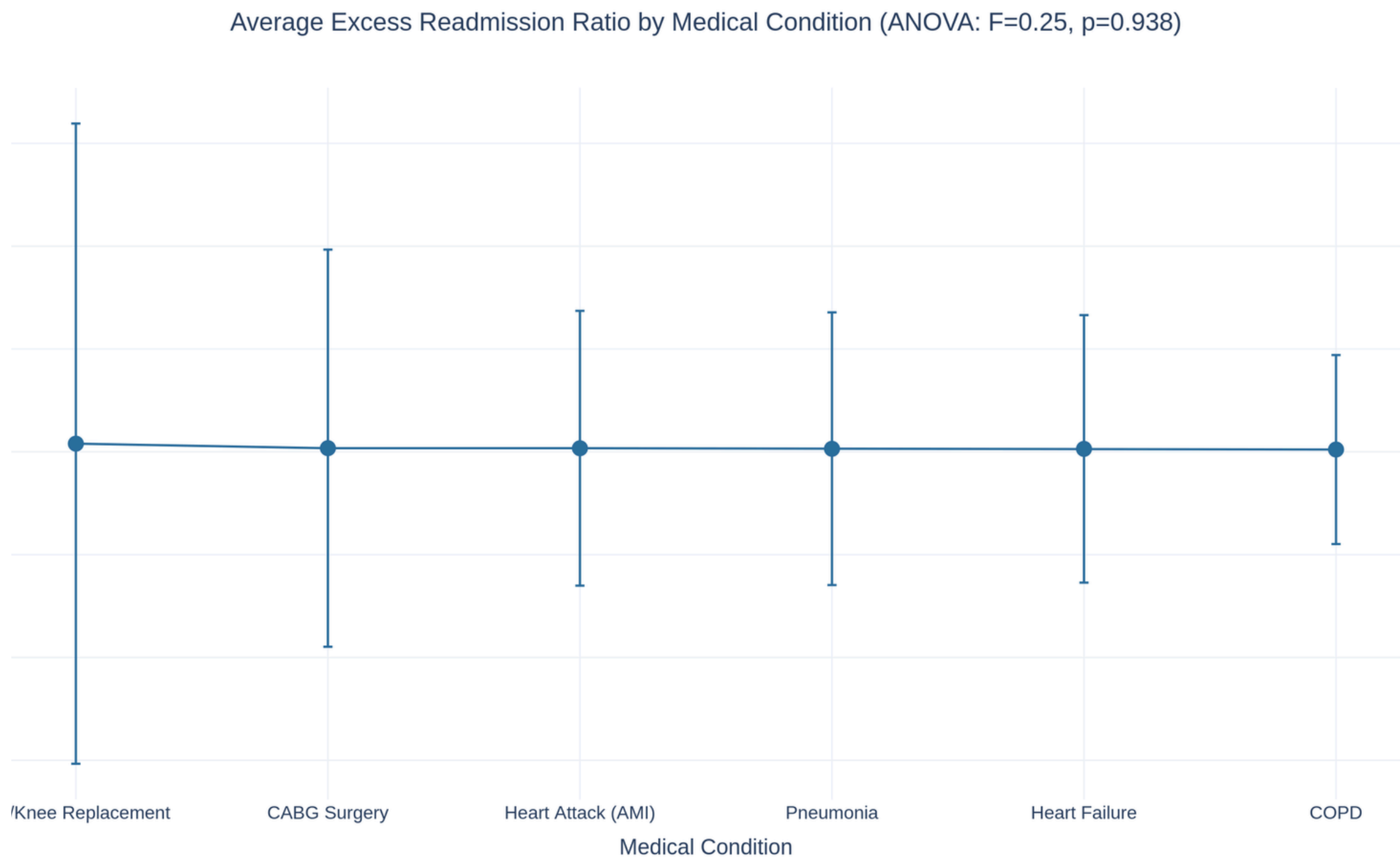
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Key Takeaway

South Dakota leads on patient experience (88.83) while keeping readmissions low (0.97). Utah, Idaho, Kansas and Colorado also land in the favorable quadrant. Notably, hospital volume doesn't predict quality several smaller states outperform larger systems on both metrics.

Which conditions are hardest on readmission performance?

One-way ANOVA: $F = 0.254$, $p = 0.938$



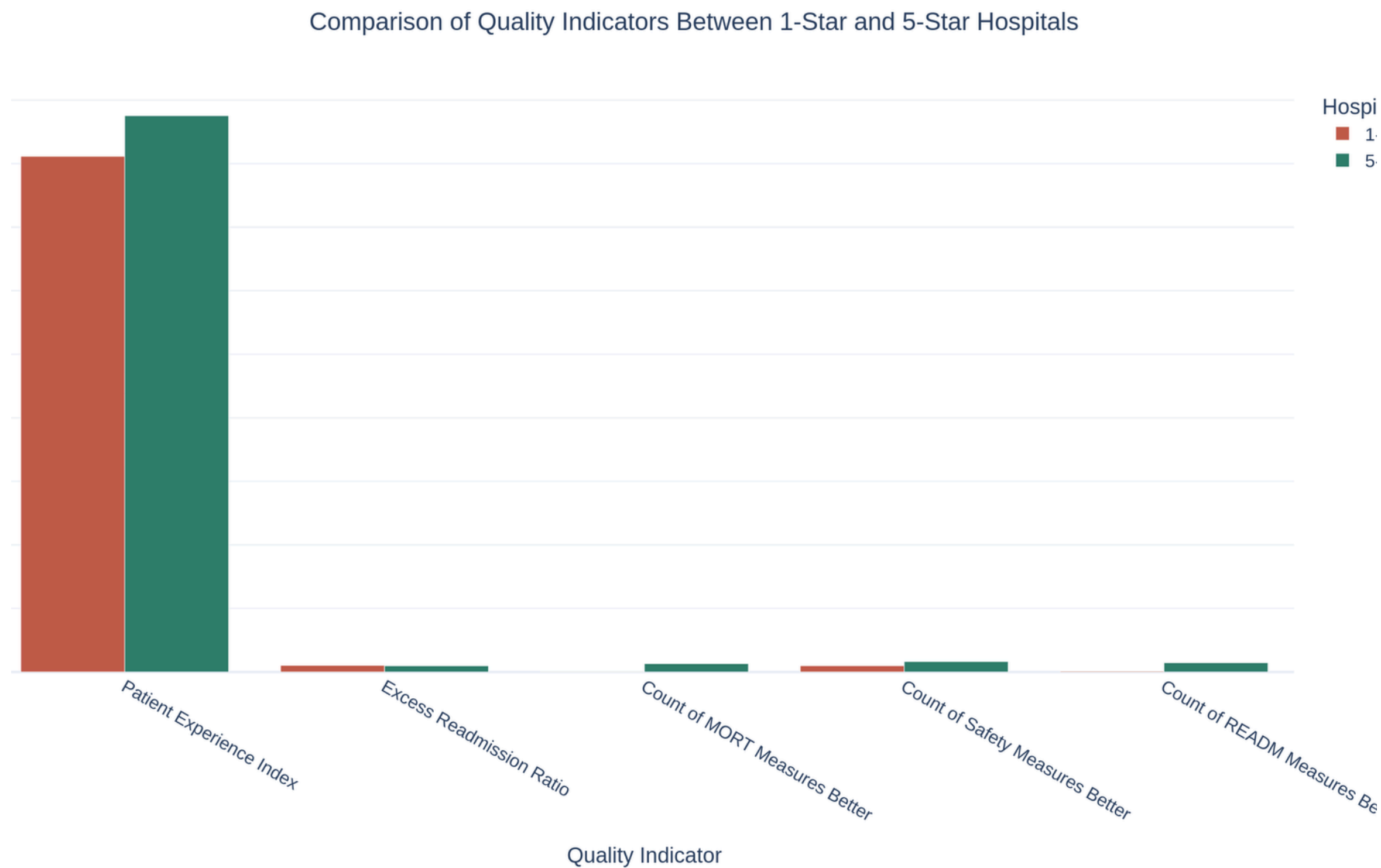
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Key Takeaway

All six conditions Hip/Knee, CABG, Heart Attack, COPD, Heart Failure, Pneumonia average close to 1.00 with no statistically significant difference between them. After CMS risk-adjustment, condition type doesn't explain performance variation; hospital-level factors matter far more.

What separates 1-star from 5-star hospitals?

Comparison across five quality indicators



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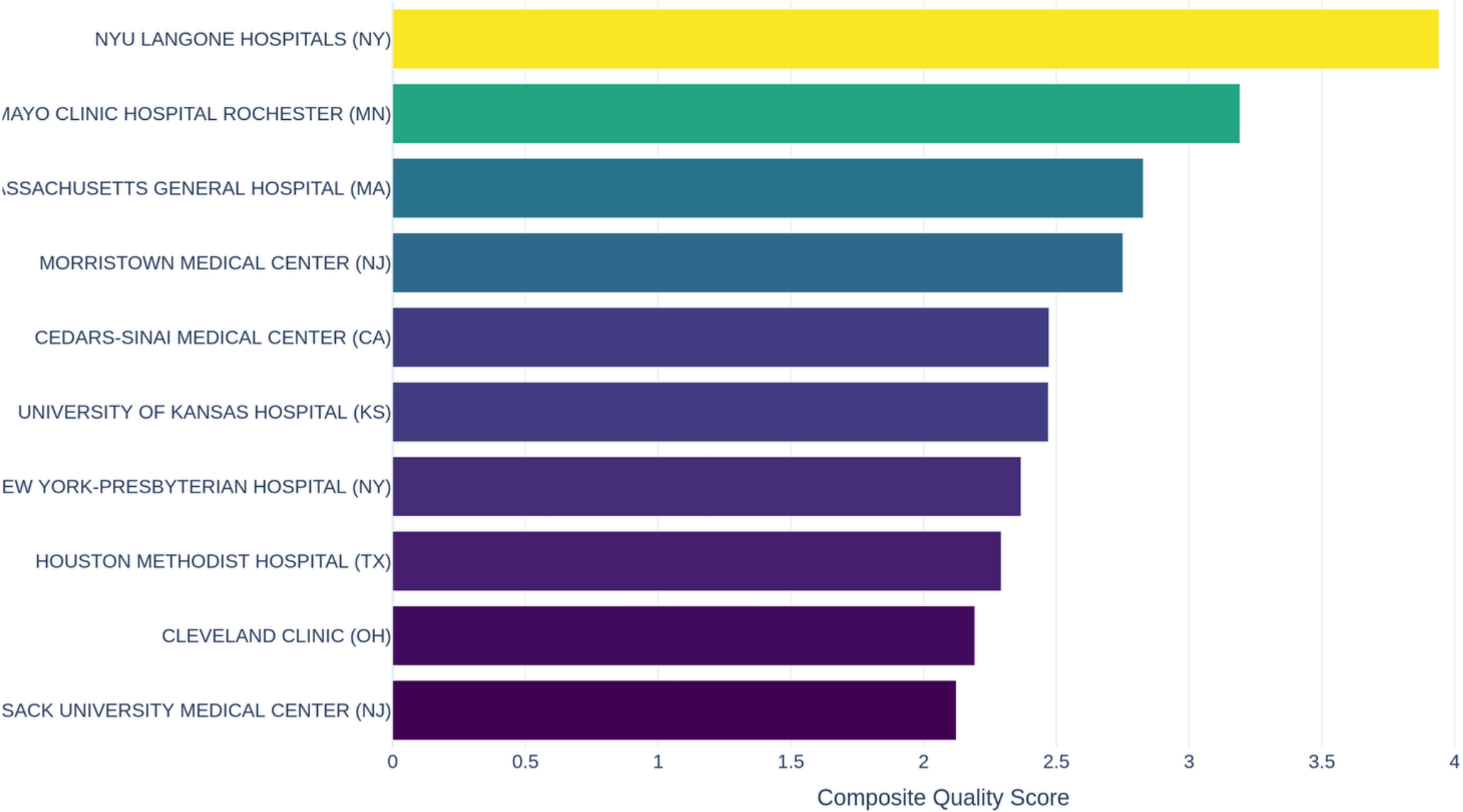
Key Takeaway

5-star hospitals beat 1-star hospitals on every dimension measured: higher Patient Experience (87.54 vs. 81.15), lower Excess Readmission Ratio (0.97 vs. 1.05), and more mortality/safety/readmission measures beating the national average. Top ratings reflect broad, consistent excellence not one strong metric.

Which hospitals are strongest across every dimension?

Composite score = standardized rating + experience + mortality + safety + readmission performance

Top 10 Hospitals by Composite Quality Score



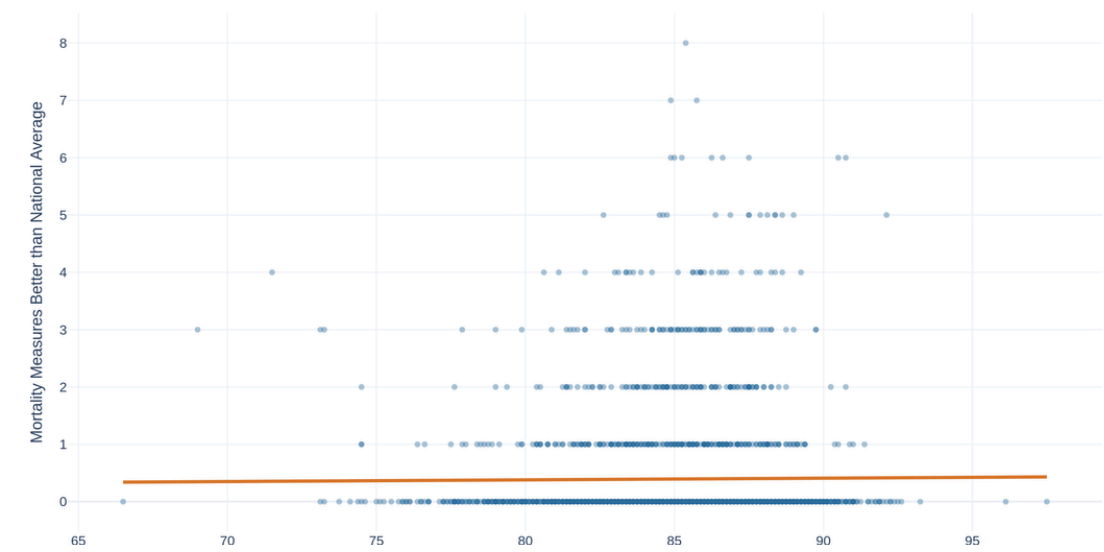
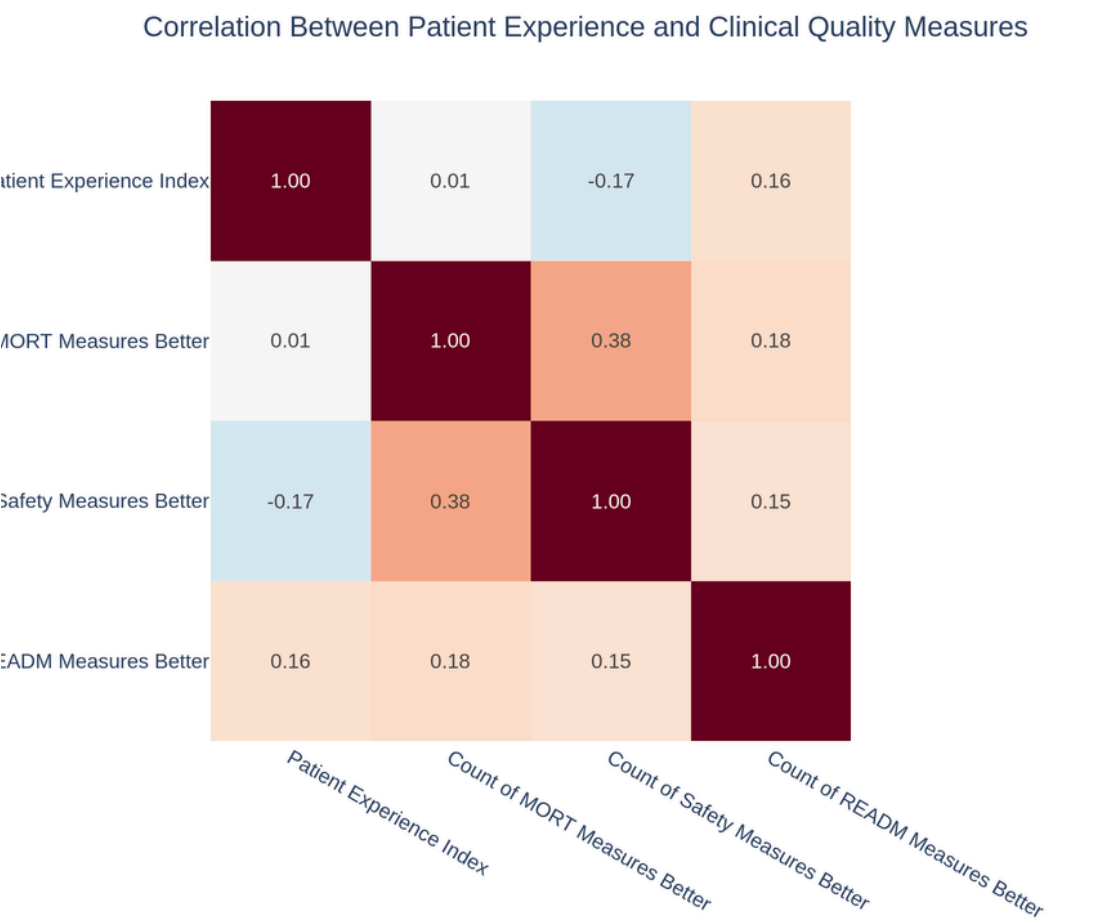
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Key Takeaway

NYU Langone Hospitals (NY) ranks #1 with a composite score of 3.94, ahead of Mayo Clinic Rochester, Massachusetts General, and Cedars-Sinai. These hospitals don't just excel at one metric they perform consistently well across patient experience, safety, and clinical outcomes simultaneously.

Do better patient experiences track with clinical quality?

Patient Experience Index vs. mortality, safety, and readmission measures



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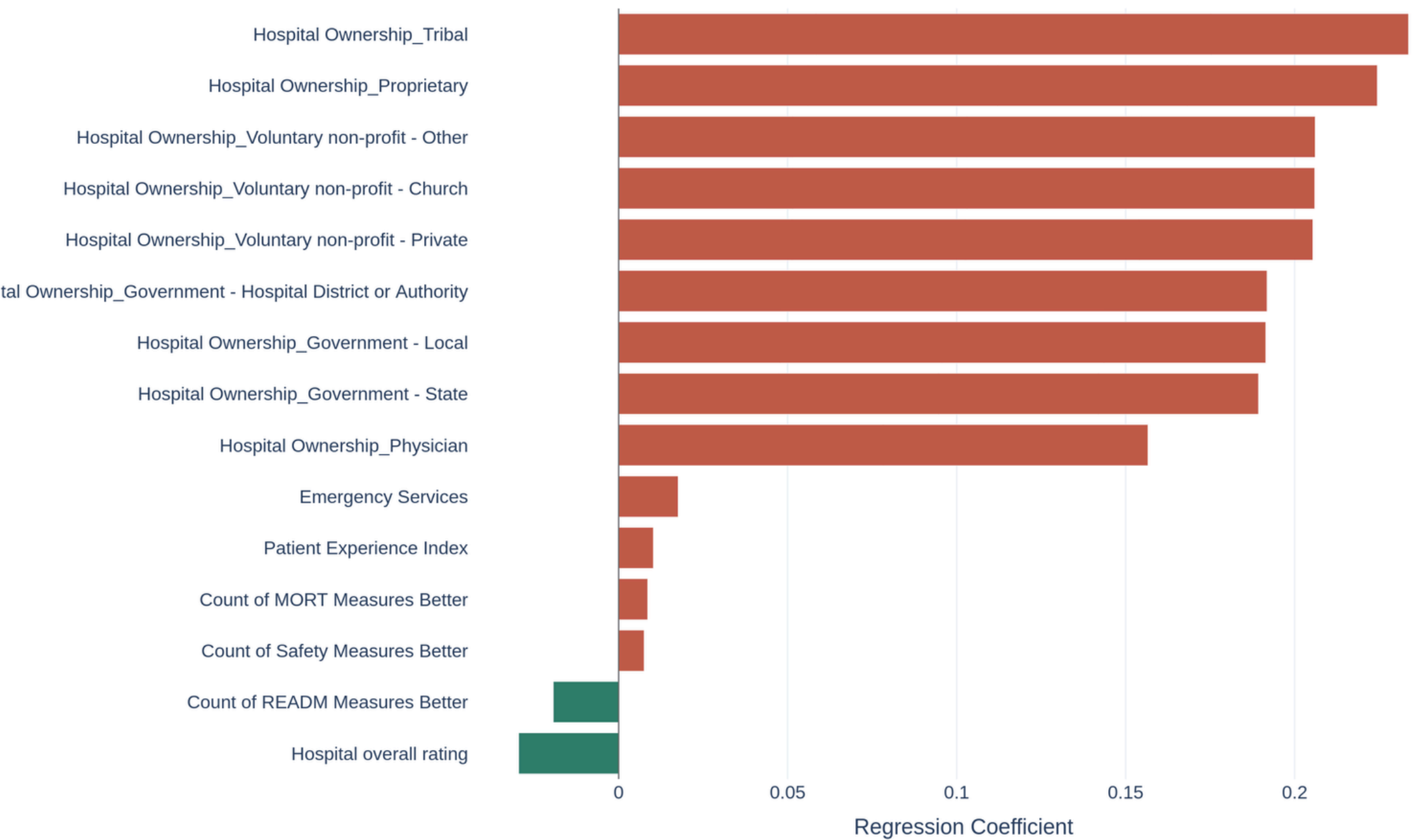
Key Takeaway

Patient experience is only weakly linked to objective clinical quality. Readmission measures show a mild positive relationship ($r = 0.16$), mortality performance is essentially unrelated ($r = 0.01$), and safety performance is weakly negative ($r = -0.17$). Satisfaction and clinical excellence are largely independent – hospitals should be judged on both, not one as a proxy for the other.

What actually drives excess readmission ratios?

Multiple OLS regression · $R^2 = 0.335$ · $F = 84.36$, $p < 0.001$

Factors Associated with Excess Readmission Ratio



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Key Takeaway

Hospital Overall Rating and Count of Readmission Measures Better are the only consistently significant negative predictors. Patient Experience, emergency-service availability, and ownership category drop out of significance once other factors are controlled for readmission performance is driven by overall hospital quality, not any single patient-facing metric.

What the Ten Questions Add Up To

Communication beats environment

Doctor communication and care transition relate to readmissions more than cleanliness or quietness.

Ratings work, but aren't the whole story

CMS overall rating predicts readmission performance - even after controlling for ownership, ER status, and experience.

Quality is multidimensional

Top hospitals (NYU Langone, Mayo Clinic, Mass General) excel across experience, safety, and outcomes together — not on one metric alone.

Experience ≠ clinical quality

Patient satisfaction is only weakly related to mortality and safety performance — both must be tracked.

Geography and condition matter less than expected

State-level quality varies (South Dakota leads), but the six clinical conditions perform nearly identically once risk-adjusted.

Explore It Yourself: The Live Dashboard

A companion Streamlit app curates six of these findings into one interactive, filterable view - explore by state, CMS rating, ownership, and emergency-service status.

- Rating vs. excess readmission ratio, filterable by state and rating
- State-level patient experience & readmission explorer
- Ownership and emergency-service comparisons
- Top 10 hospitals by composite quality score
- Regression drivers of excess readmissions



Live app URL

<https://x63yq5xuqr8wdcvo5fjr2c.streamlit.app/>

GitHub source

https://github.com/Maheboob34/maheboob_data_viz_finalproject.git

Conclusion

Hospital readmission performance is shaped by overall organizational quality not by any single lever like patient satisfaction, ownership type, or ER availability in isolation. CMS overall rating and readmission-specific quality measures are the clearest, statistically robust predictors, together explaining about a third of the variation across 3,000+ hospitals.

Limitations & next steps

- Correlational analysis -causal claims need controlled study designs
- CMS risk-adjustment may mask condition-specific dynamics worth a deeper dive
- Adding cost, staffing ratio, or socioeconomic data could sharpen the regression model

Thank you

Maheboob Patel · Data Visualization, Summer 2026